

Chapter 10

Asymptotic Analysis

We could, of course, use any notation we want; do not laugh at notations; invent them, they are powerful. In fact, mathematics is, to a large extent, invention of better notations.

— Richard Feynman, *The Feynman Lectures on Physics*, Vol. 1

In this chapter, our goal is to compare a function $f(n)$ with some simple function $g(n)$ so that we can understand its order of growth as n approaches infinity. We discuss asymptotic equality $\sim$, asymptotic tightness Θ , asymptotic upper bounds O and o , and asymptotic lower bounds Ω and ω . These asymptotic notations are widely used in computer science, mathematics, and related disciplines. These notations allow us to quickly determine the order of growth of a function without forcing us to become imprecise. We briefly illustrate the use of asymptotic notations in the analysis of algorithms.

10.1 Asymptotic Equality

Let f and g be functions from the set of positive integers to the set of real numbers. We write $f \sim g$ and say that f is **asymptotically equal** to g if and only if

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$$

holds. By definition of the limit this means that for each real number $\epsilon > 0$ there exists a positive integer n_ϵ such that

$$\left| \frac{f(n)}{g(n)} - 1 \right| < \epsilon \quad (10.1)$$

holds for all $n \geq n_\epsilon$. As the notation suggests, the positive integer n_ϵ depends on the choice of ϵ . Informally, this means that for large n , the fraction $f(n)/g(n)$ gets arbitrarily close to 1.

Another way to interpret the inequality (10.1) is that two functions f and g are asymptotically equal if and only if the relative error $(f(n) - g(n))/g(n)$ between these functions vanishes for large n . Essentially, this means that the functions f and g have the same growth for large n .

Let us see why this notation is popular, especially in analysis, combinatorics, and number theory.

Example 10.1. Suppose that we are given the function

$$f(n) = \frac{n \ln n + \sqrt{n} + \ln n + 1517 \ln \ln n}{n}.$$

If we are only interested in the growth of the function f for large n , then it is important to realize that most of the terms of f do not have a significant contribution! Indeed, the three terms

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n} = 0, \quad \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0, \quad \lim_{n \rightarrow \infty} \frac{1517 \ln \ln n}{n} = 0$$

all vanish for large n , so only the term $n \ln n / n = \ln n$ has an impact on the growth of the function $f(n)$ for large n . The asymptotic equality

$$f(n) \sim \ln n$$

expresses that $f(n)$ essentially grows like the natural logarithm $\ln n$. Since $\ln n$ is shorter and more familiar than $f(n)$, it is the preferred way to describe the behavior of $f(n)$ for large arguments n .

Another example is the Harmonic number H_n that is defined by a sum that does not have a simple closed form. The asymptotic equality nevertheless allows us to describe the growth of H_n by comparing it to a familiar function.

Example 10.2. The n -th Harmonic number $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is asymptotically equal to the natural logarithm $\ln n$,

$$H_n \sim \ln n.$$

Indeed, since the inequalities $\ln(n+1) \leq H_n \leq 1 + \ln n$ hold, dividing by $\ln n$ and taking the limit yields for the logarithmic terms

$$\lim_{n \rightarrow \infty} \frac{\ln(n+1)}{\ln n} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{1 + \ln n}{\ln n} = 1,$$

where we used l'Hôpital's rule in the calculation of the first limit. Thus, it follows from the squeeze theorem for limits that

$$\lim_{n \rightarrow \infty} \frac{H_n}{\ln n} = 1,$$

which proves that $H_n \sim \ln n$. In other words, the Harmonic numbers grow like the natural logarithm for large n . Relating a function such as H_n to the more familiar logarithmic function $\ln n$ certainly helps one to understand the growth of H_n as n approaches infinity.

Perhaps the best known example is given by Stirling's approximation, which relates the function $n!$ to a function that allows one to understand the asymptotic growth much better.

Example 10.3. The Stirling approximation of $n!$ is given by

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

The function $f(n) = n!$ grows less rapidly than n^n . Dividing n^n by e^n yields a better approximation of $n!$, but $(n/e)^n$ is always an underestimate for $n!$. By multiplication with the corrective term $\sqrt{2\pi n}$, we obtain the function $g(n) = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$ that has asymptotically the same growth as the factorial function $f(n) = n!$. The approximation works well even for relatively small numbers n . For instance, if we approximate $f(10) = 10!$ by $g(10) = \sqrt{2\pi \cdot 10} (10/e)^{10}$, then the error ϵ of the approximation is

$$\left| \frac{10!}{\sqrt{2\pi \cdot 10} \left(\frac{10}{e}\right)^{10}} - 1 \right| < \epsilon < 1.009.$$

So the multiplicative error of Stirling's approximation of $10!$ is less than 1%.

One advantage of the asymptotic equality $\sim$ is that the expression can be simplified quite a bit. The next proposition illustrates this in the case of polynomials.

Proposition 10.4. Let $p(x) = \sum_{k=0}^m a_k x^k$ be a nonzero polynomial of degree m with real coefficients. Then $p(x)$ is asymptotically equal to its leading term,

$$p(x) \sim a_m x^m.$$

Proof. By assumption, the leading coefficient $a_m \neq 0$. By expanding terms, we get

$$\lim_{x \rightarrow \infty} \frac{p(x)}{a_m x^m} = \lim_{x \rightarrow \infty} \left(1 + \frac{a_{m-1}}{a_m x} + \cdots + \frac{a_1}{a_m x^{m-1}} + \frac{a_0}{a_m x^m} \right) = 1,$$

which proves the claim. $\square$

The leading term of a polynomial dominates the growth. The advantage of the asymptotic notation $\sim$ is that we can dramatically simplify expressions such as $p(x) = x^3 + 3x^2 + 2x + 1$ when considering large x , since the relation $x^3 + 3x^2 + 2x + 1 \sim x^3$ tells us that the polynomial $p(x)$ grows like x^3 . The notation brings clarity and gives us more compact expressions. Naturally, one can derive similar propositions for many other classes of functions.

The definition of $f \sim g$ by a limit means that all the techniques that you have learned in calculus can be applied. You can apply the knowledge that you have gained in calculus to deduce asymptotic equality in situations that are not entirely obvious. As an example, we give a convenient criterion for

asymptotic equality of expressions that are of the form $f(n+k) - f(n)$ for some constant k . It is a slight generalization of a lemma given in Koecher [52, Chap. II.1.7]. The use of the mean value theorem in the proof is particularly noteworthy, since you can apply similar methods in other contexts as well.

Proposition 10.5. *Let c be a positive real number. Let f be a continuously differentiable function from the set of positive real numbers to the set of real numbers such that its derivative f' is monotonic, nonzero, and satisfies*

$$\lim_{n \rightarrow \infty} f'(n+c)/f'(n) = 1.$$

Then

$$f(n+c) - f(n) \sim cf'(n).$$

Proof. By the mean value theorem of calculus, there exists a real number θ in the range $0 \leq \theta \leq c$ such that

$$f(n+c) - f(n) = (n+c-n)f'(n+\theta) = cf'(n+\theta).$$

If f' is monotonically increasing (or monotonically decreasing), then

$$cf'(n) \underset{(\geq)}{\leq} f(n+c) - f(n) \underset{(\geq)}{\leq} cf'(n+c).$$

Dividing by $cf'(n)$ yields by assumption

$$\lim_{n \rightarrow \infty} \frac{cf'(n)}{cf'(n)} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{cf'(n+c)}{cf'(n)} = 1.$$

Therefore, by the squeeze theorem for limits, we have

$$\lim_{n \rightarrow \infty} \frac{f(n+c) - f(n)}{cf'(n)} = 1,$$

which proves our claim. □

Example 10.6. Let c be a positive constant. Then

$$\sqrt{n+c} - \sqrt{n} \sim \frac{c}{2\sqrt{n}}.$$

Indeed, if we set $f(x) = \sqrt{x}$, then f is a continuously differentiable function on the positive real numbers. Its derivative $f'(x) = 1/(2\sqrt{x})$ is nonzero, monotonically decreasing, and satisfies $\lim_{n \rightarrow \infty} f'(n+c)/f'(n) = 1$. The claim follows from Proposition 10.5.

Example 10.7. Let k be a positive real number. Then

$$\ln \left(1 + \frac{1}{n} \right)^k \sim \frac{k}{n}.$$

Indeed, choose $f(x) = k \ln(x)$. Then $f'(x) = k/x$. As $f'(x) = k/x$ is monotonically decreasing and nonzero, and

$$\lim_{n \rightarrow \infty} \frac{f'(n+1)}{f'(n)} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1,$$

it follows from Proposition 10.5 that

$$\ln \left(1 + \frac{1}{n} \right)^k = k \ln(n+1) - k \ln(n) \sim \frac{k}{n}.$$

We conclude this section by illustrating how the asymptotic equality can simplify the calculation of limits.

Proposition 10.8. *If $f_1 \sim g_1$ and $f_2 \sim g_2$, then*

$$\lim_{n \rightarrow \infty} \frac{f_1(n)}{f_2(n)} = \lim_{n \rightarrow \infty} \frac{g_1(n)}{g_2(n)}$$

provided either of these limits exist (the proviso is important!).

Proof. We multiply the limit of f_1/f_2 by 1, here written in the form $1 = (g_1 g_2)/(g_1 g_2)$. This yields

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{f_1(n)}{f_2(n)} &= \lim_{n \rightarrow \infty} \frac{f_1(n)}{g_1(n)} \frac{g_1(n)}{g_2(n)} \frac{g_2(n)}{f_2(n)} \\ &= \left(\lim_{n \rightarrow \infty} \frac{f_1(n)}{g_1(n)} \right) \left(\lim_{n \rightarrow \infty} \frac{g_1(n)}{g_2(n)} \right) \left(\lim_{n \rightarrow \infty} \frac{g_2(n)}{f_2(n)} \right), \end{aligned}$$

where we have used in the last equality that the limit of a product is the product of the limits of its factors. Since the first and last limit is 1 by hypothesis, we can conclude that

$$\lim_{n \rightarrow \infty} \frac{f_1(n)}{f_2(n)} = \lim_{n \rightarrow \infty} \frac{g_1(n)}{g_2(n)},$$

which proves the claim. $\square$

Example 10.9. Let us try to calculate the limit

$$\lim_{n \rightarrow \infty} \frac{3n^{15} + 7n^{14} + 8n^2 + \ln \sqrt{n}}{4n^{15} + 2n^7 + 2n \ln \sqrt{n}}.$$

Since the numerator and the denominator are both unbounded, we could apply l'Hôpital's rule 15 times¹ and calculate the limit. If we observe that $3n^{15} + 7n^{14} + 8n^2 + \ln \sqrt{n} \sim 3n^{15}$ and $4n^{15} + 2n^7 + 2n \ln \sqrt{n} \sim 4n^{15}$, then the previous proposition yields the limit

$$\lim_{n \rightarrow \infty} \frac{3n^{15} + 7n^{14} + 8n^2 + \ln \sqrt{n}}{4n^{15} + 2n^7 + 2n \ln \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{3n^{15}}{4n^{15}} = \frac{3}{4}$$

in a straightforward fashion.

¹Of course, there are also other ways to solve this problem. Focus on the explanation that follows!

We can use similar tricks when the numerator and denominators contain radicals.

EXERCISES

10.1. Suppose that f , g , and h are functions from the set of positive integers to the set of real numbers that are nonzero for all but a finite number of arguments. Show that

- (i) $f \sim f$,
- (ii) $f \sim g$ if and only if $g \sim f$,
- (iii) $f \sim g$ and $g \sim h$ implies $f \sim h$.

This means that $\sim$ is an equivalence relation on the set of function that are nonzero for all but a finite number of arguments.

10.2. As an important reminder that $\sim$ is an equivalence relation only on the set of functions that are *nonzero* for all but a finite number of arguments, show that there cannot be a function f such that $f \sim 0$ or $0 \sim f$.

10.3. Ernie and Bert study the functions $f(n) = n^2 + 2n$ and $g(n) = n^2$. Ernie claims that the functions are asymptotically equal, so $f \sim g$. Bert insists that f can never be asymptotically equal to g , since they always differ quite significantly, namely $f(n) - g(n) \geq 2n$ for all $n \geq 1$. Explain why one is right and why the other is wrong.

10.4. Let k and ℓ be positive integers. Show that

$$\frac{1 + n^k}{n^\ell} \sim n^{k-\ell}.$$

10.5. For a positive integer n , let $d(n)$ denote the number of divisors n . For example, $d(6) = 4$, since 6 has the divisors 1, 2, 3, and 6.

(a) Show that

$$d(n) = \sum_{k=1}^n \left(\left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right).$$

(b) The function $d(n)$ exhibits some very erratic behavior that is difficult to analyze. However, the average of the first n values of the function d is easier to analyze. Show that

$$\frac{1}{n} \sum_{k=1}^n d(k) \sim \ln n.$$

10.6. Let a be a real constant satisfying $a < 1$. Show that

$$\exp(n^a) \sim \exp((n+1)^a).$$

10.7. Show that

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}.$$

10.8.

10.9.

10.2 Limit Superior and Limit Inferior

The asymptotic equality is sometimes a bit too strict, especially when trying to work out the asymptotic behavior of sequences that fluctuate between different values. In this case, it is often better to settle for asymptotic bounds. Before embarking on the discussion of asymptotic bounds, we recall the notions of upper and lower limits from calculus. These notions turn out to be useful when trying to establish asymptotic bounds.

Upper and Lower Envelope. Let f be a function from the set of positive integers to the set of real numbers. If f oscillates between different values, then the limit $\lim_{n \rightarrow \infty} f(n)$ fails to exist. We can associate with f its upper and lower envelope, which are better behaved with respect to taking limits. The **upper envelope** u of the function f is defined by

$$u_f(n) = \sup\{f(n), f(n+1), f(n+2), \dots\},$$

and the **lower envelope** ℓ of the function f is defined by

$$\ell_f(n) = \inf\{f(n), f(n+1), f(n+2), \dots\}.$$

Example 10.10. Let $f(n)$ denote the function given by

$$f(n) = \begin{cases} 2 + 1/n & \text{if } n \text{ is even,} \\ 1 - 1/n & \text{if } n \text{ is odd.} \end{cases}$$

Then the upper envelope $u_f(n)$ of the function $f(n)$ is given by

$$\begin{aligned} u_f(1) &= \sup\{f(1), f(2), f(3), \dots\} = f(2) = 2 + 1/2, \\ u_f(2) &= \sup\{f(2), f(3), f(4), \dots\} = f(2) = 2 + 1/2, \\ u_f(3) &= \sup\{f(3), f(4), f(5), \dots\} = f(4) = 2 + 1/4, \\ u_f(4) &= \sup\{f(4), f(5), f(6), \dots\} = f(4) = 2 + 1/4, \dots \end{aligned}$$

The general term of the upper envelope $u_f(n)$ of the function f is given by

$$u_f(n) = f(2 \lfloor n/2 \rfloor) = 2 + \frac{1}{2 \lfloor n/2 \rfloor}$$

for all positive integers n .

The lower envelope $\ell_f(n)$ of the function $f(n)$ is given by

$$\begin{aligned} \ell_f(1) &= \inf\{f(1), f(2), f(3), \dots\} = f(1) = 1 - 1/1, \\ \ell_f(2) &= \inf\{f(2), f(3), f(4), \dots\} = f(3) = 1 - 1/3, \\ \ell_f(3) &= \inf\{f(3), f(4), f(5), \dots\} = f(3) = 1 - 1/3, \\ \ell_f(4) &= \inf\{f(4), f(5), f(6), \dots\} = f(5) = 1 - 1/5, \dots \end{aligned}$$

The general term of the lower envelope $\ell_f(n)$ is given by

$$\ell_f(n) = f(2 \lfloor n/2 \rfloor + 1) = 1 - \frac{1}{2 \lfloor n/2 \rfloor + 1}$$

for all positive integers n .

Figure 10.1 depicts the function f and its upper and lower envelope.

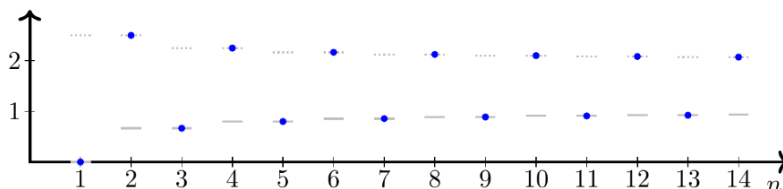


Figure 10.1: The graph shows the function (or sequence) f from Example 10.10 with values depicted by large dots. The values of the upper envelope u_f are shown by small dotted lines, and the values of the lower envelope ℓ_f by small solid lines.

A crucial point in the definition of the upper and lower envelope is that they omit more and more initial terms of f from consideration as $n \rightarrow \infty$. Therefore, the upper envelope is in general a monotonically decreasing function and the lower envelope is in general a monotonically increasing function. Consequently, the limit of the upper envelope $u_f(n)$ and the lower envelope $\ell_f(n)$ always exist in the extended real line.

Recall that the extended real line consists of the set of real numbers $\mathbf{R}$ and two symbols $-\infty$ and $+\infty$ such that $-\infty < r < \infty$ for all real numbers r . The extended real line has the convenient property that every subset has a supremum and an infimum, so it is a complete lattice. The extended real line is convenient for denoting the limit of diverging sequences. We will explicitly point out the properties related to the nonfinite values in the subsequent discussion.

Limit Inferior and Limit Superior. The **limit superior** $\limsup_{n \rightarrow \infty} f(n)$ is defined as

$$\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \left(\sup \{ f(k) \mid k \geq n \} \right)$$

and the **limit inferior** $\liminf_{n \rightarrow \infty} f(n)$ is defined as

$$\liminf_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \left(\inf \{ f(k) \mid k \geq n \} \right).$$

In other words, the limit superior is the limit of the upper envelope of f and the limit inferior is the limit of the lower envelope of f .

Let us have a look at a particularly instructive example.

Example 10.11. Let us continue the Example 10.10. Recall that the function

$$f(n) = \begin{cases} 2 + 1/n & \text{if } n \text{ is even,} \\ 1 - 1/n & \text{if } n \text{ is odd.} \end{cases}$$

has the upper envelope $u_f(n) = f(2 \lceil n/2 \rceil) = 2 + \frac{1}{2 \lceil n/2 \rceil}$ and the lower envelope $\ell_f(n) = f(2 \lfloor n/2 \rfloor + 1) = 1 - \frac{1}{2 \lfloor n/2 \rfloor + 1}$. Then

$$\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \left(2 + \frac{1}{2 \lceil n/2 \rceil} \right) = 2$$

and

$$\liminf_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2 \lfloor n/2 \rfloor + 1} \right) = 1.$$

The example shows that the limit inferior is in general not the smallest value of f , since

$$0 = 1 - 1/1 = \inf\{f(n) \mid n \geq 1\} < \liminf_{n \rightarrow \infty} f(n) = 1.$$

Likewise, the example also shows that the limit superior is in general not the largest value of f , since

$$2 = \limsup_{n \rightarrow \infty} f(n) < \sup\{f(n) \mid n \geq 1\} = 2 + 1/2.$$

We will characterize the limit inferior and limit superior as smallest and largest accumulation points of f .

The next proposition stresses the difference between the infimum and limit inferior as well as the difference between the supremum and the limit superior.

Proposition 10.12. *Let f be a function from the set of positive integers to the set of real numbers. Then*

$$\inf\{f(k) \mid k \geq 1\} \leq \liminf_{n \rightarrow \infty} f(n) \leq \limsup_{n \rightarrow \infty} f(n) \leq \sup\{f(k) \mid k \geq 1\}.$$

The previous example shows that any of these three inequalities can be strict.

Proof. The first inequality holds, since the lower envelope ℓ_f of f is a monotonically increasing function and $\ell_f(1) = \inf\{f(k) \mid k \geq 1\}$. The second inequality follows directly from the definitions. The third inequality holds, since the upper envelope u_f of f is a monotonically decreasing function and $u_f(1) = \sup\{f(k) \mid k \geq 1\}$. $\square$

Let us now discuss four different situations when the limit inferior or limit superior do not have finite values. If the function f does not have an upper bound, then the suprema $\sup\{f(n), f(n+1), f(n+2), \dots\} = \infty$ for all n , and we define

$$\limsup_{n \rightarrow \infty} f(n) = +\infty.$$

Example 10.13. Suppose that $f(n) = n$. Then $\limsup_{n \rightarrow \infty} f(n) = \infty$.

If the function f approaches $-\infty$ as $n \rightarrow \infty$, then we define

$$\limsup_{n \rightarrow \infty} f(n) = -\infty.$$

Example 10.14. Suppose that $f(n) = -n$. Then

$$u_f(n) = \sup\{-n, -(n+1), -(n+2), \dots\} = -n,$$

so $\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} u_f(n) = -\infty$.

Example 10.15. We should stress the limit superior will not be $-\infty$ when merely a subsequence approaches $-\infty$. Indeed, suppose that

$$f(n) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ -n & \text{if } n \text{ is odd.} \end{cases}$$

Then $u_f(n) = \sup\{f(n), f(n+1), f(n+2), \dots\} = 0$, so the limit superior

$$\limsup_{n \rightarrow \infty} f(n) = 0,$$

even though $f(2n+1) \rightarrow -\infty$ as $n \rightarrow \infty$.

Similarly, if the function f does not have a lower bound, then the infima $\inf\{f(n), f(n+1), f(n+2), \dots\} = -\infty$ for all n , and we define

$$\liminf_{n \rightarrow \infty} f(n) = -\infty.$$

Example 10.16. Let the function $f: \mathbf{N}_1 \rightarrow \mathbf{R}$ be given by

$$f(n) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ -n & \text{if } n \text{ is odd.} \end{cases}$$

Then $\liminf_{n \rightarrow \infty} f(n) = -\infty$ and $\limsup_{n \rightarrow \infty} f(n) = 0$.

If the function f diverges to $+\infty$ as $n \rightarrow \infty$, then we define

$$\liminf_{n \rightarrow \infty} f(n) = +\infty.$$

Example 10.17. The function $f(n) = n$ has the limit inferior

$$\liminf_{n \rightarrow \infty} n = +\infty.$$

Example 10.18. If merely a subsequence converges to $+\infty$, then the limit inferior is in general not $+\infty$. For instance, the function

$$f(n) = \begin{cases} 42 & \text{if } n \text{ is even,} \\ n & \text{if } n \text{ is odd} \end{cases}$$

has the subsequence $f(2n+1) \rightarrow \infty$ as $n \rightarrow \infty$. However, the limit inferior of $f(n)$ is given by $\liminf_{n \rightarrow \infty} f(n) = 42$.

Accumulation Points. We will now give a characterization of the finite values of the limit superior and limit inferior in terms of accumulation values. Recall that a real number a is an accumulation value of a function $f: \mathbf{N}_1 \rightarrow \mathbf{R}$ if and only if for each real number $\epsilon > 0$ there exist an infinite number of function values $f(n)$ such that

$$|f(n) - a| < \epsilon.$$

For example, if the limit $\ell = \lim_{n \rightarrow \infty} f(n)$ exists, then ℓ is an accumulation value of f . If a bounded function f does not have a limit as $n \rightarrow \infty$, then it still has accumulation points. For instance, the function f from Example 10.10 has two accumulation points, namely $a = 1$ and $a = 2$.

Let f be a function from the set of positive integers to the set of real numbers. The real number u is an **upper accumulation point** of f if and only if the following two conditions are met:

- U1.** For each real number $\epsilon > 0$ there exist infinitely many positive integers n such that $f(n) > u - \epsilon$,
- U2.** For each real number $\epsilon > 0$ there exist at most finitely many positive integers such that $f(n) > u + \epsilon$.

Evidently, an upper accumulation point is an accumulation point. If an upper accumulation point of f exists, then it is unique.

The function f is called **bounded above** if and only if there exists a real number u such that $f(n) \leq u$ holds for all positive integers n . If f has an upper accumulation point, then it is bounded above.

Proposition 10.19. *Let f be a function from the set of positive integers to the set of real numbers that is bounded above and does not diverge to $-\infty$. Then the real number u is equal to $\limsup_{n \rightarrow \infty} f(n)$ if and only if u is the upper accumulation point of f .*

Proof. ($\Rightarrow$) Suppose that the limit superior of the function f is given by the real number $u = \limsup_{n \rightarrow \infty} f(n)$. We are going to show that u must be an upper accumulation point. This means that we need to prove that (i) $f(n)$ satisfies **U1** and that (ii) $f(n)$ satisfies **U2**.

- (i) Seeking a contradiction, suppose that there exists an $\epsilon > 0$ and an n such that $f(k) \leq u - \epsilon$ for all $k \geq n$. Then $u_f(k) \leq u - \epsilon$ for all $k \geq n$, so $\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} u_f(n) \leq u - \epsilon$, contradicting our assumption $u = \limsup_{n \rightarrow \infty} f(n)$. Therefore, we can conclude that **U1** holds.
- (ii) Seeking a contradiction, suppose that there exists an $\epsilon > 0$ such that there are infinitely many positive integers n so that $f(n) > u + \epsilon$. This implies that $u_f(n) = \sup\{f(n), f(n+1), f(n+2), \dots\} > u + \epsilon$ for all $n \geq 1$. Therefore,

$$\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} u_f(n) > u + \epsilon,$$

contradicting our assumption. Consequently, for each real number $\epsilon > 0$ there exists at most finitely many positive integers such that $f(n) > u + \epsilon$, so **U2** holds.

($\Leftarrow$). Suppose that u is an upper accumulation point of f . It follows from **U1** that $u - \epsilon \leq u_f(n)$. Furthermore, it follows from **U2** that there exists a positive integer n such that $u_f(k) \leq u + \epsilon$ for all $k \geq n$. Therefore, we can conclude that for all $\epsilon > 0$, we have

$$\limsup_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} u_f(n) \in [u - \epsilon, u + \epsilon],$$

so $\limsup_{n \rightarrow \infty} f(n) = u$. $\square$

The limit superior of f can now be characterized as

$$\limsup_{n \rightarrow \infty} f(n) = \begin{cases} +\infty & \text{if } f \text{ is not bounded above,} \\ u & \text{if the upper accumulation point } u \text{ of } f \text{ exists,} \\ -\infty & \text{otherwise.} \end{cases}$$

Example 10.20. Let $f(n)$ denote the function given by

$$f(n) = \begin{cases} 2 + 1/n & \text{if } n \text{ is even} \\ 1 - 1/n & \text{if } n \text{ is odd} \end{cases}$$

For any $\epsilon > 0$, all even positive integers n such that $n > 1/\epsilon$ have a function value $f(n) > 2 - \epsilon$. We have $f(n) > 2 + \epsilon$ only when n is an even positive integer satisfying $n < 1/\epsilon$. Therefore, 2 is an upper accumulation point of $f(n)$. This is also illustrated in Figure 10.2. We can conclude that $\limsup_{n \rightarrow \infty} f(n) = 2$.

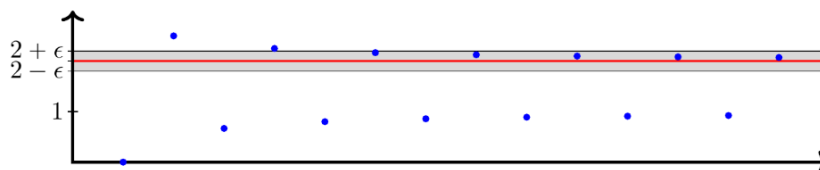


Figure 10.2: The graph shows the function (or sequence) f from Example 10.20. For each $\epsilon > 0$, there are finitely many points above $2 + \epsilon$, but an infinite number above $2 - \epsilon$, so it has an upper accumulation point of value 2. Hence $\limsup_{n \rightarrow \infty} f(n) = 2$.

The real number ℓ is called a **lower accumulation point** of f if and only if the following two conditions are met:

- L1.** For each real number $\epsilon > 0$ there exist infinitely many positive integers n such that $f(n) < \ell + \epsilon$,

L2. For each real number $\epsilon > 0$ there exist at most finitely many positive integers such that $f(n) < \ell - \epsilon$.

If a lower accumulation point of f exists, then it is unique. Figure 10.3 gives an example of the lower accumulation point for the function f from Example 10.20.

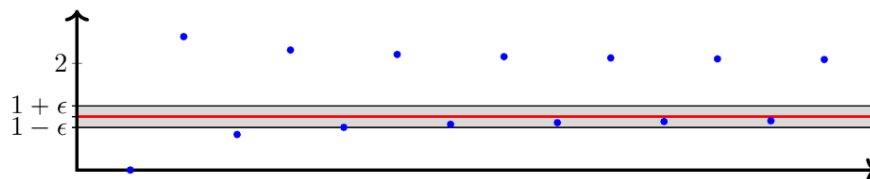


Figure 10.3: The graph shows the function (or sequence) f from Example 10.20. For each $\epsilon > 0$, there are finitely many points below $1 - \epsilon$, but an infinite number below $1 + \epsilon$, so it has a lower accumulation point of value 1. Hence $\liminf_{n \rightarrow \infty} f(n) = 1$.

The function f is called **bounded below** if and only if there exists a real number ℓ such that $f(n) \geq \ell$ holds for all positive integers n . If f has a lower accumulation point, then it is bounded below.

Proposition 10.21. *Let f be a function from the set of positive integers to the set of real numbers that is bounded below and does not diverge to ∞ . Then the real number ℓ is equal to $\liminf_{n \rightarrow \infty} f(n)$ if and only if ℓ is the lower accumulation point of f .*

Proof. Since $\liminf_{n \rightarrow \infty} f(n) = -\limsup_{n \rightarrow \infty} -f(n)$, the claim is a direct consequence of Proposition 10.19. $\square$

As a consequence of the previous proposition, the limit inferior of f can be characterized as

$$\liminf_{n \rightarrow \infty} f(n) = \begin{cases} -\infty & \text{if } f \text{ is not bounded below,} \\ \ell & \text{if the lower accumulation point } \ell \text{ of } f \text{ exists,} \\ +\infty & \text{otherwise.} \end{cases}$$

EXERCISES

10.10. Find all accumulation points of the following functions, and determine their upper and lower accumulation points.

- (a) $f(n) = (-1)^n$,
- (b) $f(n) = 4 + (-1)^n n / (n + 10)$
- (c) $f(n) = ((-1)^n + (-1)^{\lfloor n/2 \rfloor})(1 + 1/n)$.

10.11. Construct a function $f(n)$ that has all positive integers as accumulation points.

10.12. Determine $\limsup_{n \rightarrow \infty} f(n)$ and $\liminf_{n \rightarrow \infty} f(n)$ when $f(n)$ is given as

- (a) $f(n) = (-1)^n$,
- (b) $f(n) = \frac{1}{n}$,
- (c) $f(n) = (1 + (-1)^n)n$,
- (d) $f(n) = (-1)^n(2n + 1)/(n + 1)$.

10.13. Show that if $\limsup_{n \rightarrow \infty} f(n) = L$ and $\liminf_{n \rightarrow \infty} f(n) = L$, then the limit $\lim_{n \rightarrow \infty} f(n)$ exists and is equal to L .

10.14. Show that every bounded function $f: \mathbf{N}_1 \rightarrow \mathbf{R}$ has an accumulation point. This result is due to Bolzano and Weierstrass.

10.3 Asymptotically Tight Bounds

The asymptotic equality is often a bit too strict. Sometimes it is desirable to relax the constraints and consider the growth up to a constant factor and without the need for the existence of a limit. In this section, we discuss so-called asymptotically tight bounds that provide both upper and lower asymptotic bounds for a function, yet often allow one to use very simple functions to describe the growth.

Let f and g denote functions from the set of positive integers to the set of real numbers. We say that f and g have the **same order of growth** and write $f \asymp g$ if and only if there exist positive real constants c and C and a positive integer n_0 such that

$$c|g(n)| \leq |f(n)| \leq C|g(n)|$$

holds for all $n \geq n_0$. The notation $f \asymp g$ goes back to Hardy [34] and is popular in mathematics. Computer scientists like to express this in the form $f \in \Theta(g)$, where

$$\Theta(g) = \{f : \mathbf{N}_1 \rightarrow \mathbf{R} \mid f \asymp g\}$$

is the set of functions that have the same order of growth as g , see Knuth [47]. If $f \in \Theta(g)$ or $f \asymp g$, then we also say that g is an **asymptotically tight bound** for f .



In the literature, you will frequently find that $f \in \Theta(g)$ is written in the form $f = \Theta(g)$. Evidently, a function f cannot be “equal” to a set $\Theta(g)$. In the early days, the notation $\Theta(g)$ simply meant a given but unspecified representative of the set $\Theta(g)$. Even though $\Theta(g)$ is now understood as a set, the notation $f = \Theta(g)$ is commonly used instead of the more proper notation $f \in \Theta(g)$. You will need to get used to this idiosyncratic notation.

Example 10.22. As a first example, let us consider the function

$$f(n) = \frac{1}{9}n^2 + \frac{12}{10n}.$$

We compare this function with two scalar multiples of n^2 , namely $\frac{1}{10}n^2$ and $\frac{1}{8}n^2$. Then

$$\frac{1}{10}n^2 \leq f(n) \leq \frac{1}{8}n^2$$

holds for all $n \geq 5$. Indeed, the lower bound is clear. For the upper bound, it suffices to note that $n \geq 5$ implies $n^3 \geq 8 \cdot 9 \cdot 12/10$, and dividing both sides by $8 \cdot 9 \cdot n$ yields $\frac{1}{8}n^2 - \frac{1}{9}n^2 = \frac{n^2}{8 \cdot 9} \geq \frac{12}{10n}$, so adding $\frac{1}{9}n^2$ to both sides yields the upper bound $\frac{1}{8}n^2 \geq f(n)$. The graphs of these functions are shown in Figure 10.4.

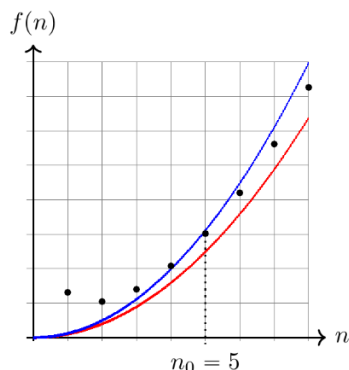


Figure 10.4: The function $f(n) = \frac{1}{9}n^2 + \frac{12}{10n}$ is shown with dots. The lower bound is given by $\frac{1}{10}n^2$ and the upper bound by $\frac{1}{8}n^2$, both shown with solid graphs even though only the values at integer arguments are relevant. The inequality $\frac{1}{10}n^2 \leq f(n) \leq \frac{1}{8}n^2$ holds for all $n \geq 5$. Therefore, $f(n) \in \Theta(n^2)$. It is okay that the first few function values $f(1), f(2), f(3)$, and $f(4)$ are not within the bounds, since the inequality is required to hold only for large arguments.

When we want to prove that $f \in \Theta(g)$ holds, we need to find constants three constants n_0 , c , and C such that

$$c|g(n)| \leq |f(n)| \leq C|g(n)|$$

holds for all $n \geq n_0$. It can be a bit cumbersome to verify this definition. The next proposition gives a convenient sufficient condition for establishing $f \in \Theta(g)$, but unfortunately it is not always applicable.

Proposition 10.23. *Let f and g be functions from the set of positive integers to the set of real numbers. If g is a positive function and the limit*

$$d = \lim_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|}$$

exists and is a nonzero real number d , then $f \in \Theta(g)$.

Proof. It follows from the definition of the limit that for each $\epsilon > 0$ there exists a positive integer n_ϵ such that

$$d - \epsilon \leq \frac{|f(n)|}{|g(n)|} \leq d + \epsilon$$

for all $n \geq n_\epsilon$. In other words, for the constants $c = d - \epsilon$ and $C = d + \epsilon$ there exists an n_ϵ such that $c|g(n)| \leq |f(n)| \leq C|g(n)|$ holds for all $n \geq n_\epsilon$, which proves $f \in \Theta(g)$. $\square$

If f and g are asymptotically equal, then their limit of their quotient is 1. We record this simple observation in the next corollary.

Corollary 10.24. *If two functions f and g are asymptotically equal, $f \sim g$, then they have the same order of growth, so $f \asymp g$.*

Example 10.25. Recall that the expected number of comparisons in randomized Quicksort is given by $2(n+1)H_n - 4n$. Since $H_n \sim \ln n$, it follows that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2(n+1)H_n - 4n}{n \ln n} &= \lim_{n \rightarrow \infty} \frac{2(n+1) \ln n - 4n}{n \ln n} \\ &= \lim_{n \rightarrow \infty} \frac{2n \ln n + 2 \ln n - 4n}{n \ln n} = 2. \end{aligned}$$

It follows from Proposition 10.23 that the expected number of comparisons of Quicksort is given by $\Theta(n \ln n)$. Even though the expression $2(n+1)H_n - 4n$ is more exact, it is probably better to use the more straightforward bound $\Theta(n \ln n)$ when communicating to fellow programmers.

The next example shows that even when the limit of the quotient of two functions does not exist, then they might still have the same order of growth.

Example 10.26. Let $f(n) = (2 + (-1)^n)n^2$ and $g(n) = n^2$. Then the limit

$$\lim_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|}$$

does not exist, as the quotient fluctuates between 3 and 1, but

$$|g(n)| \leq |f(n)| \leq 3|g(n)|$$

holds for all $n \geq 1$; hence, $f \in \Theta(g)$.

We call a function f from the set of positive integers to the set of real numbers **eventually nonzero** if and only if there exists an integer n_0 such that $f(n)$ is nonzero for all $n \geq n_0$.

The next proposition gives a necessary and sufficient condition for asymptotically tight bounds.

Proposition 10.27. *Let f be a function from the set of positive integers to the set of real numbers, and g an eventually nonzero function from the set of integers to the set of real numbers. Then $f \in \Theta(g)$ if and only if*

$$\liminf_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} > 0 \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} < \infty.$$

Proof. If $f \in \Theta(g)$, then there exists a positive constants c and C and a positive integer n_0 such that $c \leq |f(n)|/|g(n)| \leq C$ holds for all $n \geq n_0$. This implies that

$$\liminf_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} \geq c > 0 \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} \leq C < \infty$$

hold.

Conversely, suppose that f and g are functions satisfying

$$c := \liminf_{n \rightarrow \infty} |f(n)|/|g(n)| > 0 \quad \text{and} \quad C := \limsup_{n \rightarrow \infty} |f(n)|/|g(n)| < \infty.$$

By definition of the limit superior and inferior, for any ϵ in the range $0 < \epsilon < c$ there exists a positive integer n_0 such that

$$0 < c - \epsilon \leq \frac{|f(n)|}{|g(n)|} \leq (C + \epsilon)$$

holds for all $n \geq n_0$. Multiplying these inequalities by $|g(n)|$ shows that $f \in \Theta(g)$ holds. $\square$

Example 10.28. On input of a positive integer n , a program executes a block of code $T(n)$ times, where

$$T(n) = \begin{cases} 2^n & \text{when } n \text{ is even,} \\ 2^{n+10+(\log_2 n)/n} & \text{when } n \text{ is odd.} \end{cases}$$

We claim that $T(n) \in \Theta(2^n)$. Indeed, let us calculate the limit inferior and limit superior of $T(n)/2^n$. We have

$$(T(1)/2^1, T(2)/2^2, T(3)/2^3, \dots) = (2^{10+\log_2(1)}, 1, 2^{10+\log_2(3)/3}, 1, 2^{10+\log_2(5)/5}, 1, 2^{10+\log_2(7)/7}, \dots).$$

In other words, the quotient $T(n)/2^n = 1$ when n is even and $T(n)/2^n = 2^{10+(\log_2 n)/n}$ when n is odd. Therefore, we get

$$\liminf_{n \rightarrow \infty} \frac{T(n)}{2^n} = 1 \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{T(n)}{2^n} = 2^{10}.$$

Since the limit inferior and the limit superior are both positive real numbers, it follows from Proposition 10.27 that $T(n) \in \Theta(2^n)$.

A characteristic of an asymptotically tight bound $f \in \Theta(g)$ is that $f(n)$ cannot fluctuate too much in comparison to $g(n)$. In general, we want to keep the functions g simple. For instance, $g(n) = n^k$ for some positive integer k , $g(n) = b^n$ for some positive integer b , or similar elementary functions. A given function f might not be as nicely behaved as the simple functions that we use for comparison. Then we might be forced to use separate simple functions as upper and lower asymptotic bounds. We describe the appropriate asymptotic bounds in the next sections.

EXERCISES

10.15. Suppose that f, g , and h are functions from the set of positive integers to the set of real numbers. Show that

- (i) $f \asymp f$,
- (ii) $f \asymp g$ if and only if $g \asymp f$,
- (iii) $f \asymp g$ and $g \asymp h$ implies $f \asymp h$.

In other words, show that $\asymp$ is an equivalence relation.

10.16. Show that a polynomial $p(x)$ of degree m satisfies $p(x) = \Theta(x^m)$.

10.17. Let $f(n) = 2n^3 + 3n + 2$ and $g(n) = 5n^4 + 3n^3 + 2n + 1$. Express $\Theta(fg)$ in the simplest possible terms.

10.18. Let b and d be positive real numbers that are not equal to 1.

- (a) Show that $\Theta(\log_b n) = \Theta(\log_d n)$, so one can write $\Theta(\log n)$ using a baseless logarithm without causing confusion.
- (b) Prove or disprove: Does $\Theta(n^{\log_b n}) = \Theta(n^{\log_d n})$ hold in general?

10.19. Let a be a real number, $a > 1$. Show that

$$\sum_{k=0}^n a^k = \Theta(a^n).$$

10.20. Show that for all positive integers k , we have

$$1^k + 2^k + \cdots + n^k = \Theta(n^{k+1}).$$

10.4 Asymptotic Upper Bounds

The asymptotic equality and asymptotic tight bounds classify functions of similar growth. In this and the next section, we are concerned with relating functions of different growths.

Let f and g be functions from the set of positive integers to the set of real numbers. We say that g is an **asymptotic upper bound** for f and write $f \in O(g)$ if and only if there exists a positive real constant C and a positive integer n_0 such that

$$|f(n)| \leq C|g(n)|$$

holds for all $n \geq n_0$.

We denote by $O(g)$ the set of all functions $f: \mathbf{N}_1 \rightarrow \mathbf{R}$ such that g is an asymptotic upper bound for f .



The relation $f \in O(g)$ is often written as $f = O(g)$ in the literature.

Even worse, subset relations such as $O(n) \subseteq O(n^2)$ are often denoted as $O(n) = O(n^2)$. These idiosyncratic notations are here to stay, so you need to get used to them. If you read statements in the literature such as $\ln n = O(n) = O(n^2)$, simply translate the equalities into the usual “element of” and “subset relations”, such as $\ln n \in O(n) \subseteq O(n^2)$.

The asymptotic upper bound relation $f \in O(g)$ allows us to compare the growth of functions for large arguments n . Figure 10.5 shows the graphs of three functions. While it is possible to gain some insights into the growth of functions, it is easy to be misled by such graphs, as they show the values for the functions for some sets of small values of n . Yet the asymptotic upper bound is concerned with the behavior of the functions at large values of their arguments. For this reason, it is preferable to give a proof for the bounds.

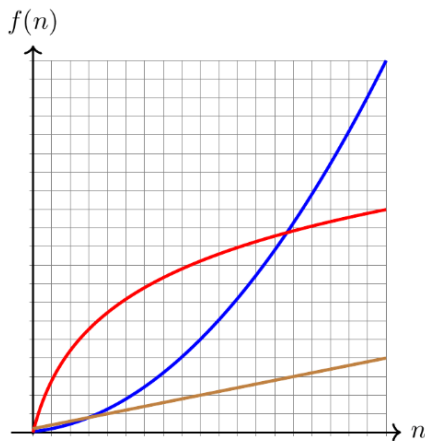


Figure 10.5: The graphs of the functions $\frac{1}{5}n$, $4 \ln n$, and $\frac{1}{20}n^2$ on the interval $[1, 20]$. The quick growth of the function $\frac{1}{20}n^2$ suggests that $\frac{1}{5}n \in O(\frac{1}{20}n^2)$ and $4 \ln n \in O(\frac{1}{20}n^2)$. However, graphs are not a good tool to judge such relationships. For example, $4 \ln n$ exceeds $\frac{1}{5}n$ throughout this graph, but it is not difficult to show that $4 \ln n \in O(\frac{1}{5}n)$, whereas $\frac{1}{5}n \notin O(4 \ln n)$.

Example 10.29. We have $n \in O(n^2)$. Indeed, for all integers $n \geq 1$, multiplying both sides by n yields $n \leq n^2$. Therefore, we have

$$|n| \leq 1|n^2|.$$

for all $n \geq 1$. The same argument shows that $|n| \leq 1|-n^2|$, so $n \in O(-n^2)$. For positive functions, there is no need to use absolute values.

Example 10.30. Suppose that k and ℓ are integers such that $1 \leq k < \ell$. For all integers $n \geq 1$, raising both sides to the $(\ell - k)$ -th power yields $n^{\ell - k} \geq 1$. If we then multiply both sides by n^k , we get $n^\ell \geq n^k$. Therefore, we can conclude that $n^k \in O(n^\ell)$.

The next proposition gives a convenient necessary and sufficient condition to establish an asymptotic upper bound $f \in O(g)$.

Proposition 10.31. *Let f be a function from the set of positive integers to the set of real numbers, and g an eventually nonzero function from the set of positive integers to the set of real numbers. Then $f(n) \in O(g(n))$ if and only if*

$$\limsup_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} < \infty.$$

Proof. The proof is straightforward, but instructive if you are not yet familiar with the notion of the limit superior, see Exercise 10.23. $\square$

Corollary 10.32. *Let f and g be functions from the set of positive integers to the set of real numbers. If the limit*

$$\lim_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|}$$

exists and is finite, then $f \in O(g)$.

Example 10.33. Let us consider the functions $\ln n$ and n . We claim that $\ln n \in O(n)$. Since both $\ln n \rightarrow \infty$ and $n \rightarrow \infty$ as $n \rightarrow \infty$, we can use l'Hôpital's rule to deduce that

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Therefore, it follows from Corollary 10.32 that $\ln n \in O(n)$, as claimed.

Figure 10.6 gives some rules that are convenient when trying to establish asymptotic upper bounds. For example, it follows from the constant rule that $2n \in O(2n) = O(n)$. You can gain familiarity with asymptotic upper bounds by proving these rules.

Let f and g be functions from the set of positive integers to the set of real numbers. We say that g is a **strict asymptotic upper bound** for f and write $f \in o(g)$ if and only if for every real number $\epsilon > 0$ there exists a positive integer n_ϵ such that

$$|f(n)| \leq \epsilon |g(n)|$$

holds for all $n \geq n_\epsilon$. By definition, $f \in o(g)$ implies that $f \in O(g)$.

If $f \in o(g)$, then f is asymptotically much smaller than g . This property is nicely explained by the next proposition.

Proposition 10.34. *Let f and g be functions from the set of positive integers to the set of real numbers such that g is eventually nonzero. Then $f \in o(g)$ if and only if*

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \tag{10.11}$$

holds.

Constants. If c is a nonzero constant, then

$$cO(f(n)) = O(f(n)), \quad (10.2)$$

$$O(cf(n)) = O(f(n)). \quad (10.3)$$

Idempotency. The Big Oh operator is idempotent, meaning that

$$O(O(f(n))) = O(f(n)). \quad (10.4)$$

Multiplications. The multiplication of Big Oh expressions follows the rules

$$O(f(n))O(g(n)) = O(f(n)g(n)), \quad (10.5)$$

$$O(f(n)g(n)) = f(n)O(g(n)). \quad (10.6)$$

Absorption. We can simplify Big Oh expressions using the rule

$$O(f(n)) + O(g(n)) = O(g(n)) \text{ provided that } f(n) \in O(g(n)). \quad (10.7)$$

Powers. For all positive integers k , we have

$$(f(n) + g(n))^k \in O((f(n))^k) + O((g(n))^k). \quad (10.8)$$

Linear Combinations. If $f(n) \in O(h(n))$ and $g(n) \in O(h(n))$, then

$$af(n) + bg(n) \in O(h(n)) \text{ for all } a, b \in \mathbf{R}. \quad (10.9)$$

Swap. The next rule allows you to swap Big Oh terms.

$$\text{If } f(n) \in g(n) + O(h(n)) \text{ then } g(n) \in f(n) + O(h(n)). \quad (10.10)$$

Figure 10.6: Some simple rules for the manipulation of upper bounds.

Proof. Suppose that (10.11) holds. By definition of the limit, this means that for any $\epsilon > 0$ there exists a positive integer n_ϵ such that

$$\left| \frac{f(n)}{g(n)} \right| < \epsilon$$

holds for all $n \geq n_\epsilon$. This is equivalent to the condition that for each $\epsilon > 0$ there exists an n_ϵ such that

$$|f(n)| \leq \epsilon |g(n)|$$

holds for all $n \geq n_\epsilon$. In other words, (10.11) is equivalent to $f \in o(g)$. $\square$

Corollary 10.35. Let f and g be functions from the set of positive integers to the set of real numbers. Suppose that $f \in o(g)$. Then

$$g + f \in O(g).$$

Example 10.36. We claim that $n \ln n \in o(n^2)$, so $n \ln n \in O(n^2)$. Indeed, it follows from l'Hôpital's rule that

$$\lim_{n \rightarrow \infty} \frac{n \ln n}{n^2} = \lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Therefore, we can deduce from Proposition 10.34 that $n \ln n \in o(n^2)$.

EXERCISES

10.21. Let $f(n) = 4 \ln n$, $g(n) = \frac{1}{5}n$, and $h(n) = \frac{1}{20}n^2$. Show that

- (a) $f \in o(g)$,
- (b) $g \in o(h)$.
- (c) Conclude that $f \in O(g)$ and $g \in O(h)$.

10.22. Let f and g be positive functions. Show that $f \in o(g)$ implies that $g \notin O(f)$.

10.23. Prove Proposition 10.31: Let f be a function from the set of positive integers to the set of real numbers, and g an eventually nonzero function from the set of positive integers to the set of real numbers. Then $f(n) \in O(g(n))$ if and only if

$$\limsup_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} < \infty.$$

10.24. Prove the idempotency rule given by equation (10.4).

10.25. Prove the multiplication rules (10.5) and (10.6).

10.26. Prove the absorption rule (10.7) and the power rule (10.8).

10.27. Prove the linear combination rule given in (10.9).

10.28. Prove the rule (10.10).

10.29. Find at least five examples of an erroneous usage of the Big Oh notation in recent papers or books of Computer Science.

10.30. Suppose that $p(n) = a_0 + a_1n + \cdots + a_m n^m$ is a polynomial of degree m with complex coefficients. Show that $p(n) = O(n^k)$ for all $k \geq m$, and $p(n) \neq O(n^\ell)$ for $0 \leq \ell < m$.

10.31. Show that for fixed k , we have

$$\binom{n}{k} \in \frac{n^k}{k!} + O(n^{k-1}) \quad \text{and} \quad \binom{n+k}{k} \in \frac{n^k}{k!} + O(n^{k-1}),$$

where $\binom{n}{k} = n(n-1)(n-2) \cdots (n-k+1)/k!$ is the binomial coefficient.

10.32. Show that $n/(n+1) \in 1 + O(1/n)$.

10.33. Let f, g , and h be functions from the set of positive integers to the set of real numbers. We write $f \leq g$ if and only if $f \in O(g)$. Show that

- (i) $f \leq f$
 - (ii) $f \leq g$ and $g \leq h$ implies $f \leq h$
 - (iii) there exist functions f and g such that neither $f \leq g$ nor $g \leq f$ holds.
- In the language of set theory, properties (i) and (ii) mean that $\leq$ is a preorder. The third property shows that the asymptotic growth of functions is not always comparable.

10.34. Prove or disprove: $e^n \in O(2^n)$.

10.35. Prove or disprove: $n^{\ln n} \in O(e^{(\ln n)^2})$.

10.36. Show that $f \in O(g)$ and $f \notin o(g)$ does not imply that $f \in \Theta(g)$.

10.37. Show that $f(x) \in g(x) + O(1)$ implies that $\exp(f(x)) \asymp \exp(g(x))$.

10.5 Asymptotic Lower Bounds

In this section, we define asymptotic lower bounds, which complement the notions that we have defined in the previous section.

Let f and g be functions from the set of positive integers to the set of real numbers. We say that g is an **asymptotic lower bound** to f and write $f \in \Omega(g)$ if and only if there exists a positive constant c and a positive integer n_0 such that

$$c|g(n)| \leq |f(n)|$$

holds for all $n \geq n_0$. This formalizes the notion that $f(n)$ grows at least as fast as a constant multiple of $g(n)$ for large n .

This asymptotic lower bound is related to the asymptotic upper bound in the following way.

Proposition 10.37. *Let f and g be functions from the set of positive integers to the set of real numbers. We have $f \in \Omega(g)$ if and only if $g \in O(f)$.*

Proof. We have $f \in \Omega(g)$ if and only if there exists a positive constant c and a positive integer n_0 such that $c|g(n)| \leq |f(n)|$ holds for all $n \geq n_0$. Dividing both sides by c shows that there exist a positive constant $C = 1/c$ and a positive integer n_0 such that $|g(n)| \leq \frac{1}{c}|f(n)| = C|f(n)|$ holds for all $n \geq n_0$. However, this is nothing but the definition of $g \in O(f)$. $\square$

Let f and g be functions from the set of positive integers to the set of real numbers. We say that g is a **strict asymptotic lower bound** to f and write $f \in \omega(g)$ if and only if for all positive constants c there exists a positive integer n_0 such that

$$c|g(n)| \leq |f(n)|$$

holds for all $n \geq n_0$.

Example 10.38. The function n^2 is in $\omega(n)$, since for a given positive constant c , the inequality $cn = c|n| \leq |n^2| = n^2$ holds for all positive integer $n \geq c$. On the other hand, n is not in $\omega(n)$, since there does not exist any positive integer n for which $2n = 2|n| \leq |n| = n$ holds.

Proposition 10.39. *Let f and g be functions from the set of positive integers to the set of real numbers, and assume that g is eventually nonzero. Then we have $f \in \omega(g)$ if and only if*

$$\lim_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} = \infty.$$

Proof. We have $f \in \omega(g)$ if and only if for all positive constants c there exists a positive integer n_0 such that $c \leq |f(n)|/|g(n)|$ holds for all $n \geq n_0$, so $|f(n)|/|g(n)|$ grows without bound. By definition of the limit, this is equivalent to

$$\lim_{n \rightarrow \infty} \frac{|f(n)|}{|g(n)|} = \infty,$$

which proves the claim. □

10.6 Analysis of Algorithms

In this section, we will give a brief exposition of the analysis of algorithms. The purpose of the analysis is to get some insights into the running time or the space requirements of the computation. Since too many details of the compiler and computer architecture affect the precise time or space requirements, one has to settle for the more modest goal of obtaining reasonable bounds on these quantities. We will focus on the analysis of the running time of an algorithm.

Since we are settling for bounds on the running time rather than a precise estimate, we can use a simplified model of computation such as the random access machine. A random access machine is a very simple single-processor machine that executes instructions one after another without pipelining, speculative execution, branch prediction, or parallelism. It even lacks a memory hierarchy. We could now precisely define the instructions of the random access machine and their costs, as in Aho, Hopcroft, and Ullman [1]. However, we follow the approach taken in Cormen, Leiserson, Rivest, and Stein [16] and assume that elementary operations take a constant – yet unspecified – amount of time. Then there is no need to translate the pseudocode into random access machine instructions, which simplifies matters.

The running time of an algorithm is measured as a function of the size of the input. What is the size of the input? Unfortunately, the answer to this question depends on the application. We assume that we are given a function s that associates to an input x its **size** $s(x)$. For instance, if the input to a sorting algorithm is an array x with n elements, then $s(x) = n$ would be a natural measure for the size of the input. If the input x is an integer, then $s(x) = \lfloor \log_{10} x \rfloor + 1$ may be a good measure of input size for algorithms that deal

with long integer arithmetic. If the integers remain bounded, then $s(x) = 1$ would be a reasonable choice.

Let I denote the set of all possible inputs to an algorithm A . If we denote by $t_A(x)$ the running time of the algorithm A on an input x , then the **worst-case running time** of the algorithm A is defined as

$$\sup \{t_A(x) \mid x \in I, s(x) = n\}.$$

This measures the worst-case performance of the algorithm for all inputs of length n . We will use asymptotic notations to simplify the resulting worst-case running time.

So how do we determine the worst-case running time of an algorithm in practice? The answer is that we simply analyze the algorithm line-by-line and essentially count operations. The main complications are conditional statements, loops, and function or procedure calls. We now put forth the axioms that govern the run-time analysis. The list is incomplete, but you will see the pattern and it should be easy to extend this list by further language constructs.

A1. Elementary operations have constant running time.

Examples of elementary operations include assignments, arithmetic with machine size words, Boolean operations, comparisons, and array access, among others. The axioms says that they all have $\Theta(1)$ running time.

The running time of a compound statement is bounded by the sum of the running times of the individual component statements.

A2. If the statements S_1 and S_2 respectively have worst case running time $T(S_1)$ and $T(S_2)$ on an input of size n , then the compound statement $S_1; S_2$ has worst-case running time

$$T(S_1; S_2) \leq T(S_1) + T(S_2)$$

The next axiom gives a coarse-grained view of the running-time of a conditional statement. We should add that sometimes it makes sense to distinguish cases depending on the value of the condition C .

A3. The conditional statement

if C then S_1 else S_2

has worst-case running time $T(C) + \max\{T(S_1), T(S_2)\}$ when the conditional statement C and the statements S_1 and S_2 have worst case running time $T(C)$, $T(S_1)$ and $T(S_2)$ on an input of size n , respectively.

The next axiom gives the estimate for the running-time of a for loop, which iterates through all integers k in the range $a \leq k \leq b$, where a and b are integers. The notation $(a..b)$

A4. The running-time of a for-loop statement S given by

```

for  $k$  in  $(a..b)$  do
   $S_1$ 
end
```

has worst-case running time $T(S) \leq c + \sum_{k=a}^b (T(S_1, k) + c)$, where $T(S_1, k)$ is the worst-case running time of statement S_1 during iteration k and c is the cost of evaluating the loop condition.

In the worst case, such a for loop iterates $b - a + 1$ times and checks the loop condition $b - a + 2$ times (as the loop verifies that k is out of bounds before it stops the iteration). One special case is worth noting. If $T(S_1) = T(S_1, k)$ is nonzero and independent of k , then this simplifies to

$$T(S) \leq (b - a + 1)\Theta(T(S_1)) + (b - a + 2)c.$$

There exist numerous variations of for loops. We will leave it to the reader to modify the axiom for these variations.

The running-time of a do-while loop (sometimes also called a repeat-until loop) is given in the next axiom. This style of loop executes the statement S_1 at least once before checking the condition C . It will repeat while the condition C is met.

A5. The running-time of a do-while loop statement S given by

```

begin
   $S_1$ 
end while  $C$ 
```

has worst-case running time $T(S) \leq \sum_{k=1}^{m(n)} (T(C) + T(S_1, k))$, where $m(n)$ is the maximal number of loop iterations on an input of size n and $T(C)$ and $T(S_1)$ are the worst-case running times of the conditional statement C and the statement S_1 , respectively.

We illustrate how to obtain an estimate for the worst-case running-time with a small example.

Example 10.40. Suppose that you want to write a program to evaluate a polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

so for a given input x , you would like to know the value $y = p(x)$. You can save multiplications by using the Horner scheme

$$p(x) = (\cdots ((a_n x + a_{n-1})x + a_{n-2})x + \cdots a_1)x + a_0.$$

The following algorithm is based on the Horner scheme:

horner(a, n, x)	cost	times
res = a[n]	c_1	1
for k = n-1 down to 0 do	c_2	n+1
res = res * x + a[k]	c_3	n
end	c_4	n
return res	c_5	1
end	c_6	1

The worst-case running time of the algorithm is given by

$$T(n) = c_1 + c_2(n + 1) + c_3n + c_4n + c_5 + c_6 = \Theta(n).$$

This compares favorably with the naive $\Theta(n^2)$ algorithm that evaluates the polynomial $p(x)$ by computing each power independently by repeated multiplication.

EXERCISES

10.38. Analyze the running time of the following algorithm using a step count analysis (as in the Horner scheme).

```
// search a key in an array a[1..n] of length n
search(a, n, key)                                cost  times
for k in (1..n) do
  if a[k]=key then
    return k
end
return false
```

Determine the worst-case complexity of this algorithm.

10.39. Analyze the running time of the following algorithm using a step count analysis (as in the Horner scheme).

```
// determine the number of digits of an integer num
binary_digits(num)                                cost  times
int cnt = 1
while (num>1) do
  cnt = cnt + 1
  num = floor(num/2.0)
end
return cnt
```

- (a) Give an axiom for the running time of a while loop.
- (b) Determine the asymptotic running-time of this algorithm.

10.40. Determine how many times the block **Block(a,b)** is executed in the following code fragment:

```

n = 100
for a in (0..n-1) do
  for b in (0..n) do
    Block(a,b)
  end
end

```

You can assume that **Block(a,b)** does not exit the for loops, nor does it manipulate the loop variables **a** and **b**.

10.41. Determine how many times the block **Block(a,b)** is executed in the following code fragment:

```

n = 100
for a in (0..n) do
  for b in (0..a) do
    Block(a,b)
  end
end

```

You can assume that **Block(a,b)** does not exit the encapsulating for loops, and does not manipulate the loop variables **a** and **b**.

10.7 Notes

Asymptotic notations were introduced by Bachmann [7] and further popularized by Landau [55]. Graham, Knuth, and Patashnik [30] give a more in-depth treatment of asymptotic notations. Spencer [71] dedicates an entire book to asymptotic notations and gives numerous surprising applications. Books on the analysis of algorithms contain numerous examples of asymptotic estimates of the running time, see for example Aho, Hopcroft, and Ullman [1] or Cormen, Leiserson, Rivest, and Stein [16].